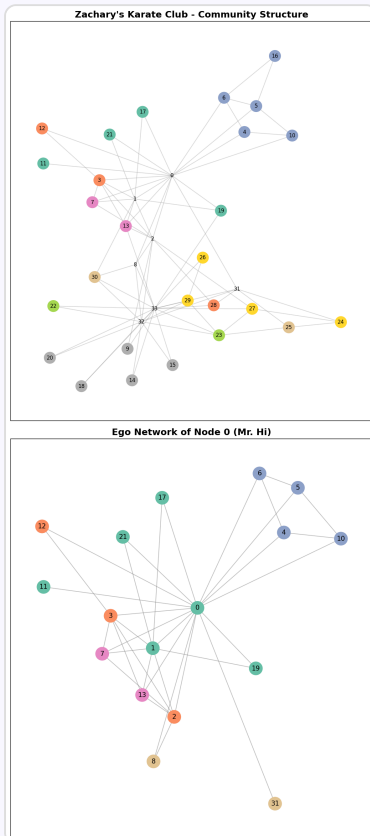


Network Visualization

Graph Analysis with NetworkX

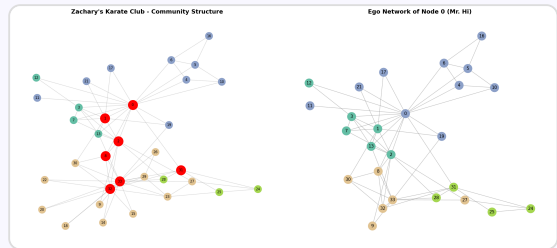
Your goal: You have a script that visualizes Zachary's Karate Club social network with community detection. The output has several issues — missing nodes, wrong layout, too many communities, and visible axis frames. Fix these to match the expected result.

Your Starting Point



Run `code.py` to generate this

Your Expected Result



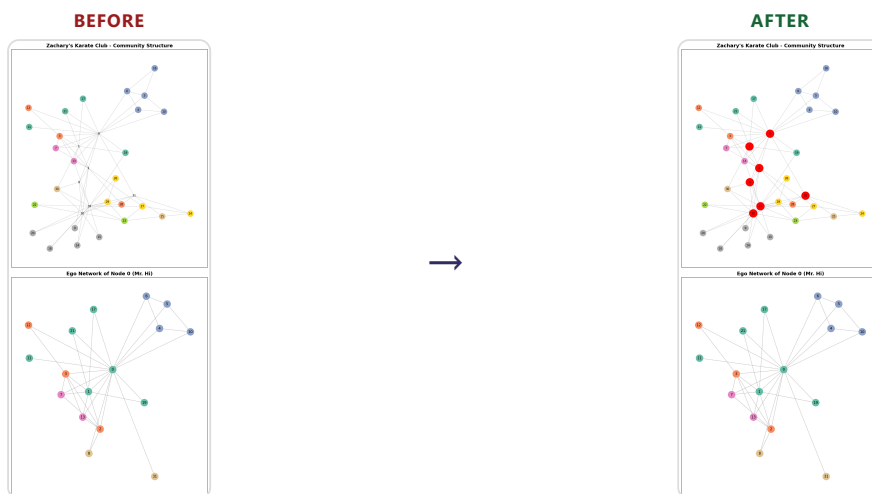
Note: Node positions may vary slightly between runs — that's normal. Focus on getting the right community colors, all nodes visible, correct layout, and clean presentation.

Edit `code.py` — complete the 4 subtasks below. Compare your output to the expected result as you go.

Subtasks

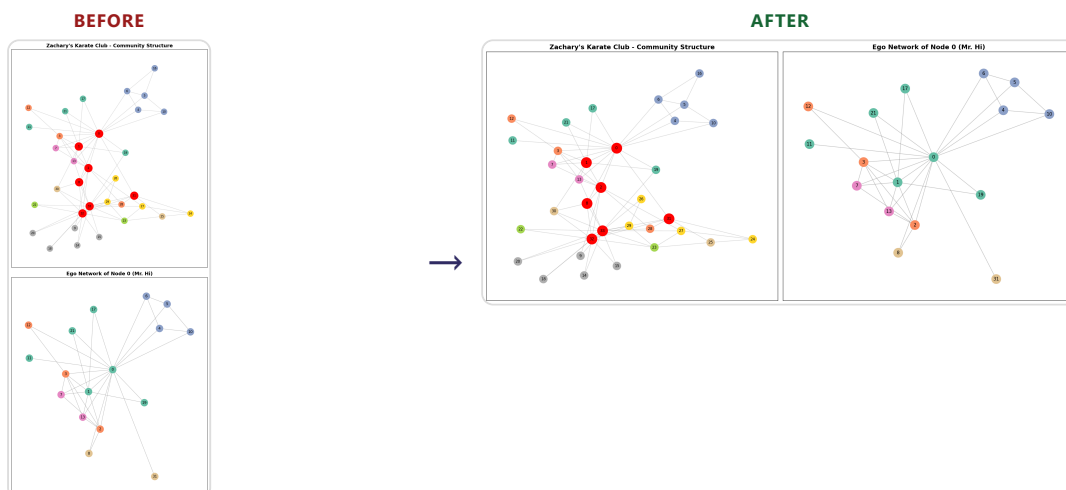
1 Draw the missing bridge nodes

High-centrality "bridge" nodes that connect communities are not being rendered. Add them to the plot with a larger size (500) and a distinct red color so they stand out.



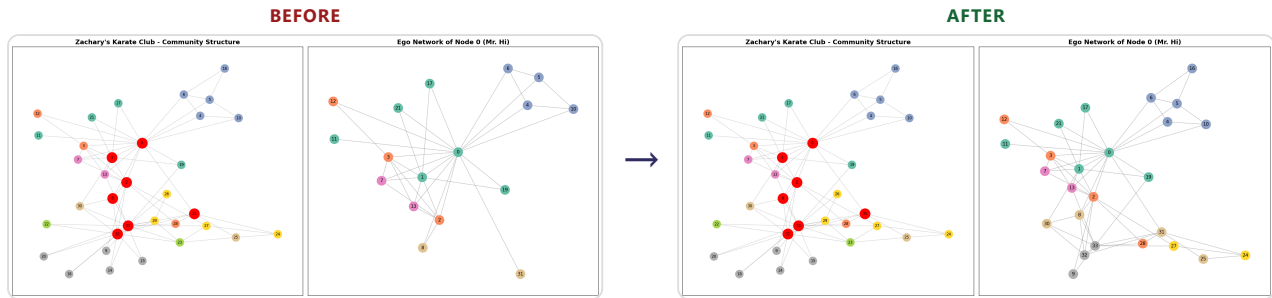
2 Place plots side by side

The full network and ego network are stacked vertically. Arrange them side by side (1 row, 2 columns) and adjust the figure size.



3 Expand the ego network

The ego network of node 0 only shows direct neighbors. Expand it to include neighbors-of-neighbors to reveal more structure.



4 Fix communities and remove axis frames

The resolution parameter is too high, splitting the network into too many tiny communities. Lower it to get the expected 2–4 groups. Also, turn off axis frames on both plots.



Edit `code.py` to make these changes. Compare your output to `III-Your-Expected-Result.png` as you go.